



Application Note

Post Processing, Angle and INL Calculation

for Analog Output Signals

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1 Introduction

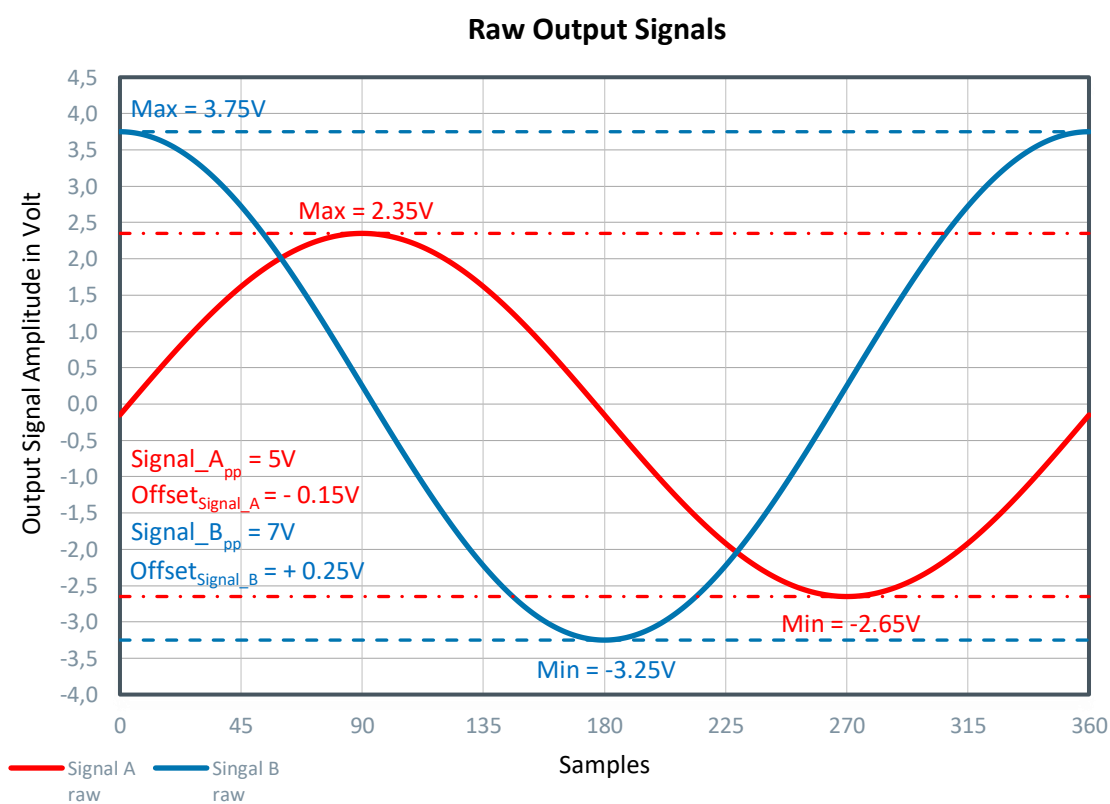
Depending on the application, it sometimes occurs that the analog output signals of a sensor IC have different amplitudes and are afflicted by an offset. The purpose of this document is, to give an overview of how to normalize gain and offset of analog output signals. Furthermore, it provides information, of how to calculate an angle and the corresponding INL out of two sinusoidal shaped signals.

2 Gain and Offset Correction

2.1 Raw Output Signals

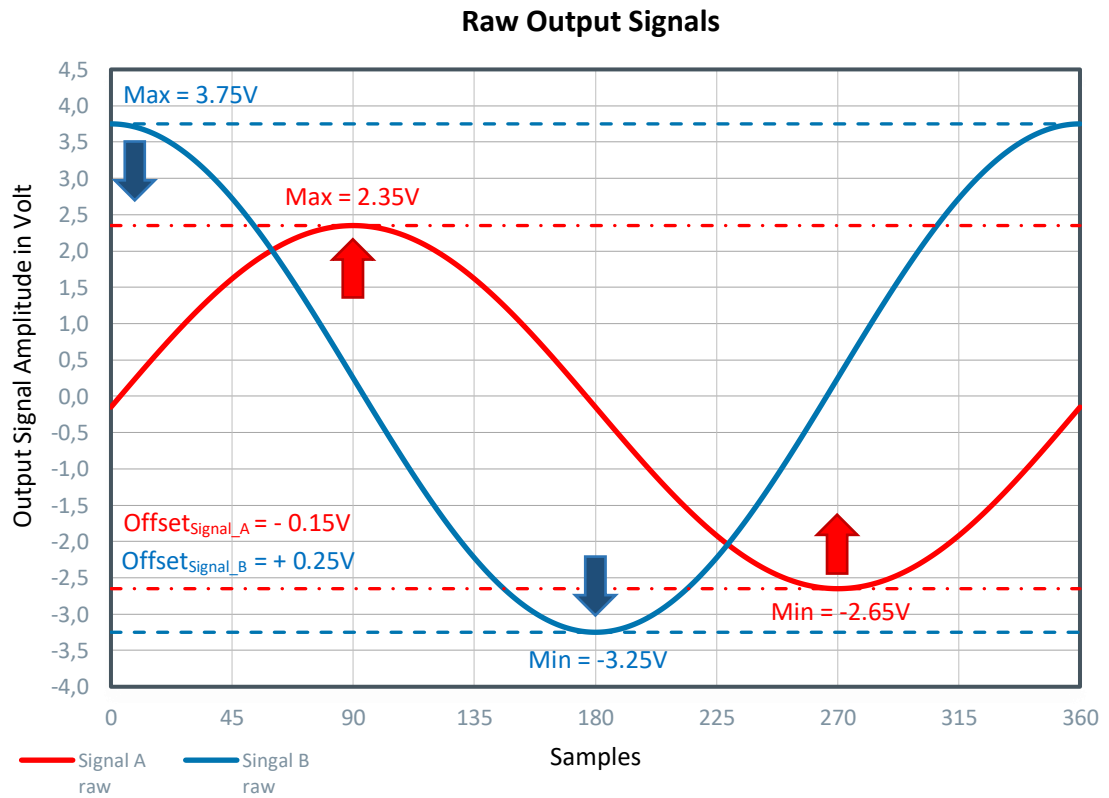
Figure 1 shows two exemplary analog output signals, Signal A and Signal B, which have different amplitudes and are afflicted by an offset.

Figure 1:
Raw Output Signals



2.2 Offset Correction

Figure 2:
Offset Correction



The purpose of the offset correction is, to equalize the positive and negative amplitudes of the signals. The following two steps accomplish this.

2.2.1 Offset Calculation

The offset is calculated out of the output signals minimum and maximum values according the following formula.

Equation 1: Offset Calculation

$$Offset = \frac{Signal_{Max} + Signal_{Min}}{2}$$

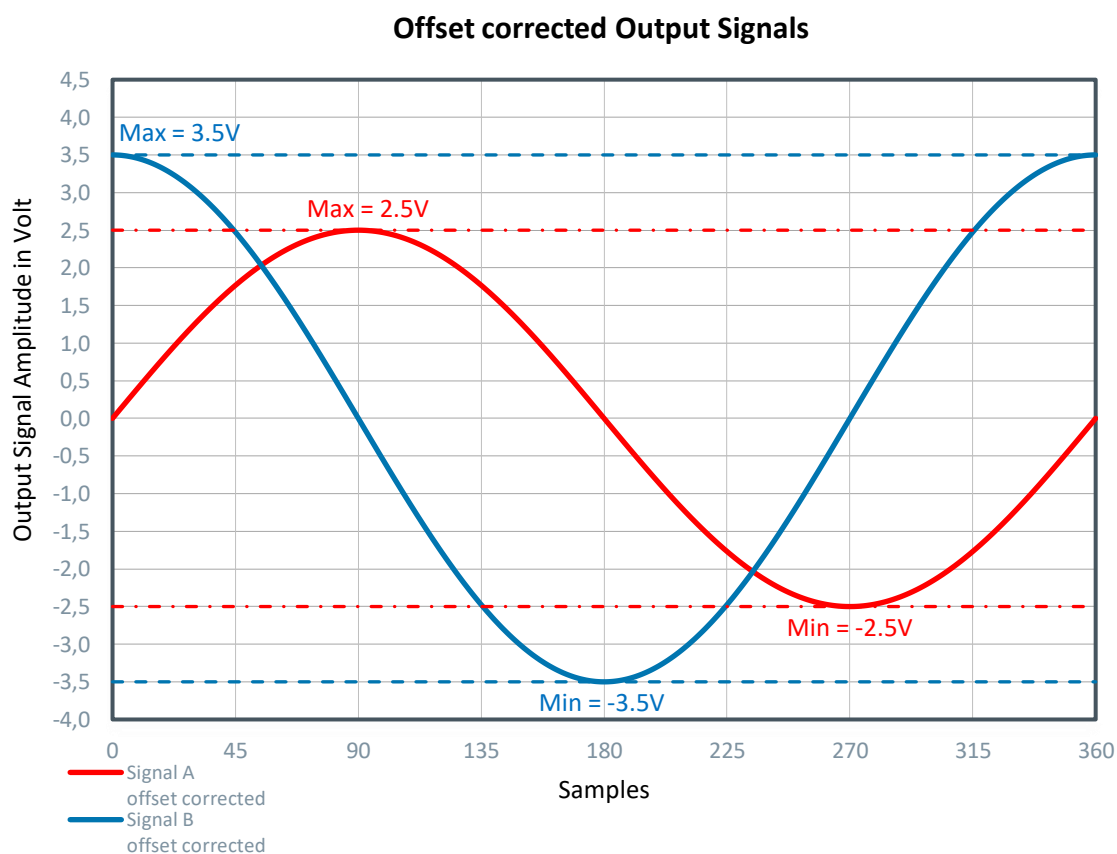
2.2.2 Offset corrected Output Signal

An offset corrected signal is obtained, by subtracting the previously calculated offset from the raw output signal.

Equation 2: Offset Correction

$$Signal_{offset_corrected} = Signal_{Raw} - Offset$$

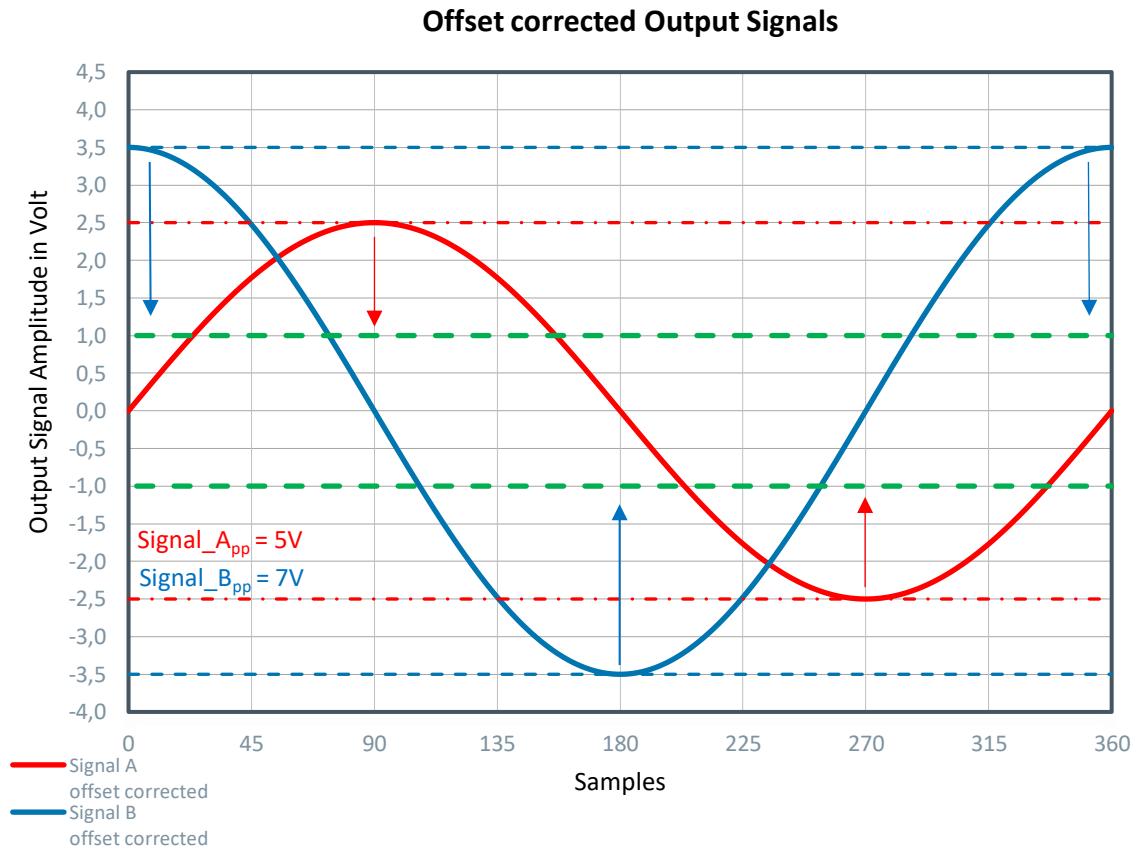
Figure 3:
Offset corrected Output Signals



As shown in Figure 3, the positive and negative peaks of each signal are equalized, but their individual amplitudes are still different.

2.3 Gain Correction

Figure 4:
Gain Correction



The purpose of the gain correction is, to equalize the amplitudes of both output signals. Furthermore, the amplitudes are normalized to 1V.

2.3.1 Gain corrected Output Signals

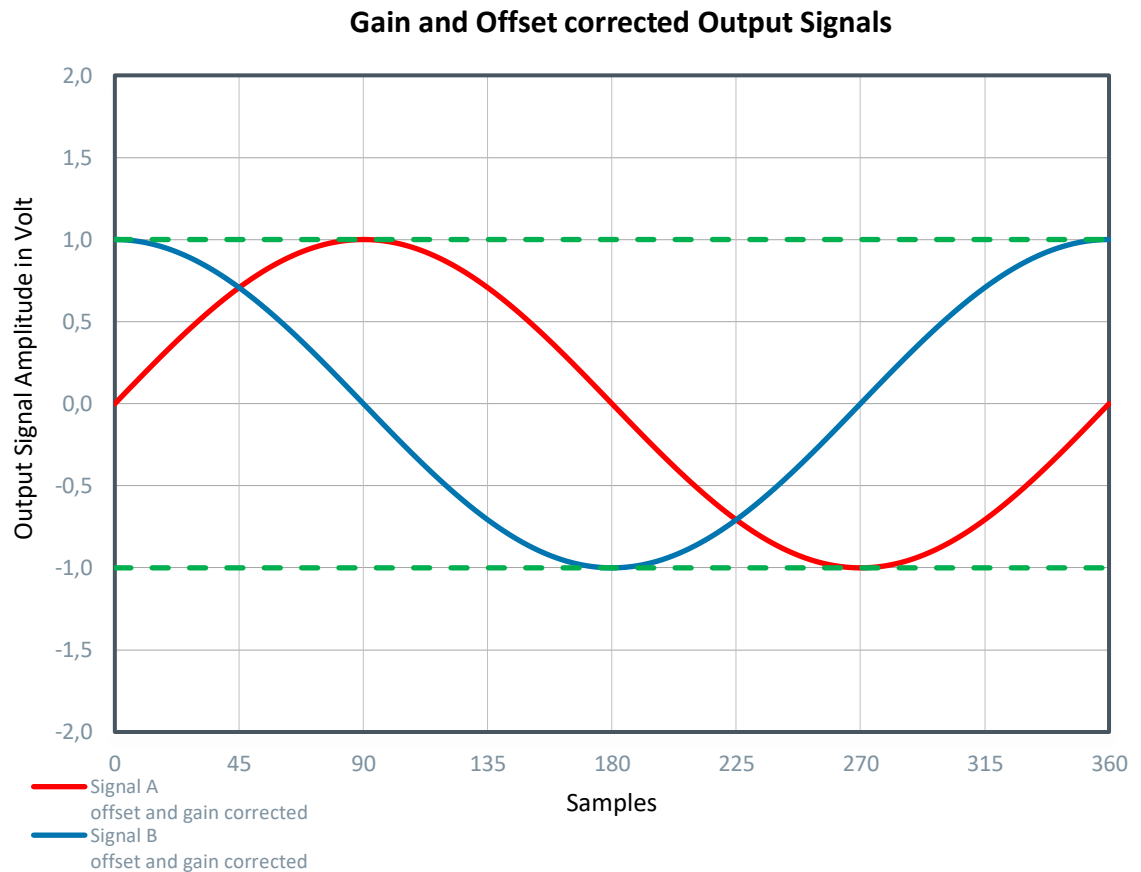
The following two formulas describe how the normalization of the output signals is carried out.

Equation 3: Gain Correction for Signal A

$$Signal_{A_{gain_corrected}} = \frac{Signal_A}{\frac{Signal_{A_{pp}}}{2}}$$

Equation 4: Gain Correction for Signal B

$$Signal_B_{gain_corrected} = \frac{Signal_B}{\frac{Signal_B_{pp}}{2}}$$

Figure 5: Gain and Offset corrected Output Signals

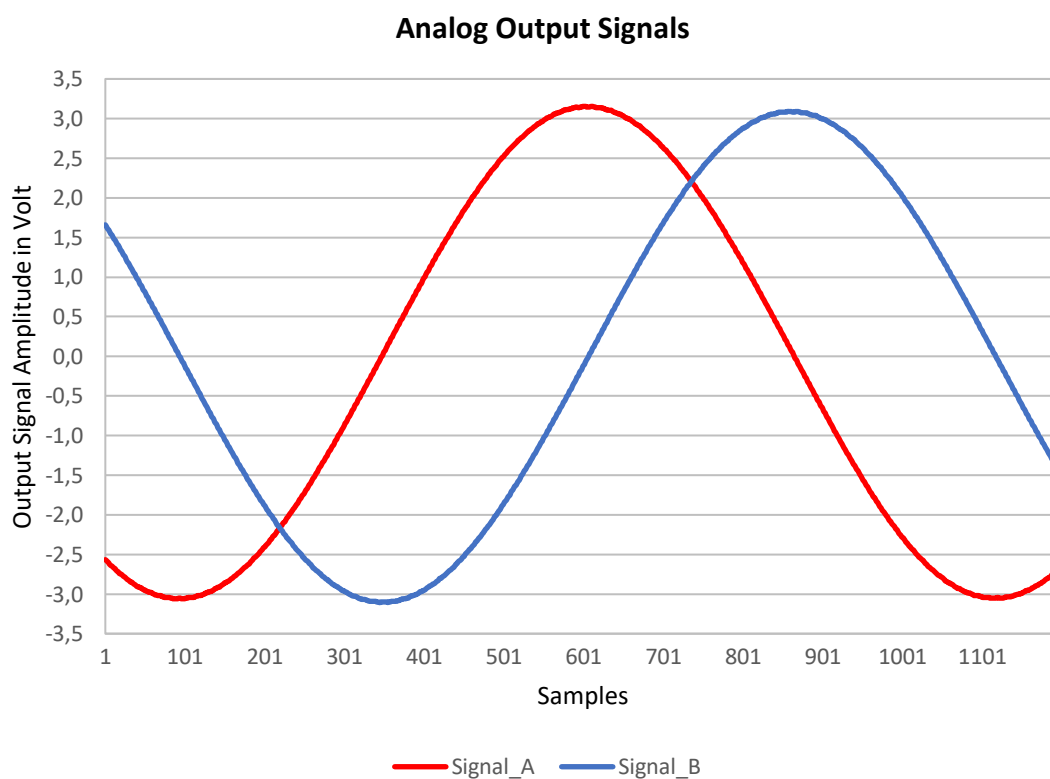
Now both signals do have the same amplitude and are free of any offset.

3 Angle Calculation

The following chapter explains how to calculate an angle out of two sinusoidal shaped signals.

3.1 Exemplary Output Signals

Figure 6:
Analog Output Signals



The diagram above shows two exemplary analog output signals.

3.2 Angle Calculation with atan2() function

To calculate an angle out of two sinusoidal shaped signals, the mathematical function `atan2()` is used. It is an angle function, which is part of most programming languages mathematic libraries. The `atan2()` function is also implemented in Microsoft Excel.

Equation 5: Angle Calculation with atan2()

$$ATAN2(\text{Signal_A}; \text{Signal_B}) = \varphi \text{ [rad]}$$

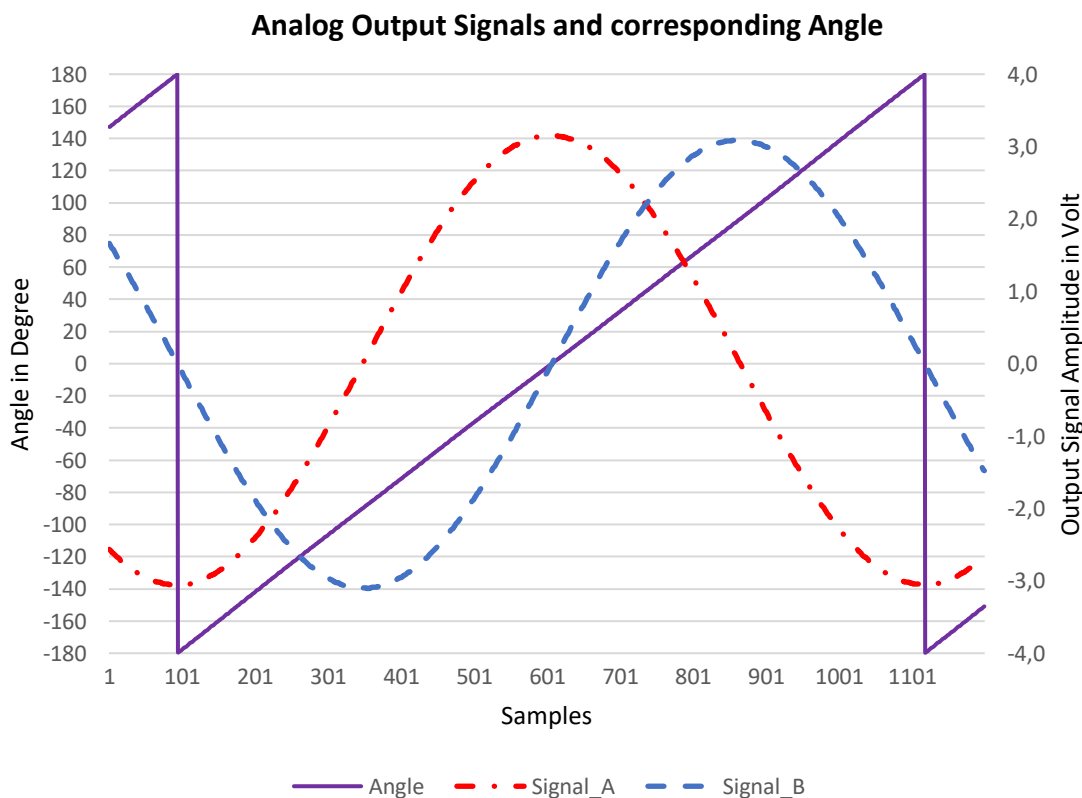
The result of the atan2() function is gives in radians. With the following formula the result is converted to degree.

Equation 6: Radians to Degree

$$\varphi \text{ [deg]} = \varphi \text{ [rad]} * \left(\frac{180}{\pi}\right)$$

3.3 Resulting Angle

Figure 7:
Analog Output Signals and corresponding Angle



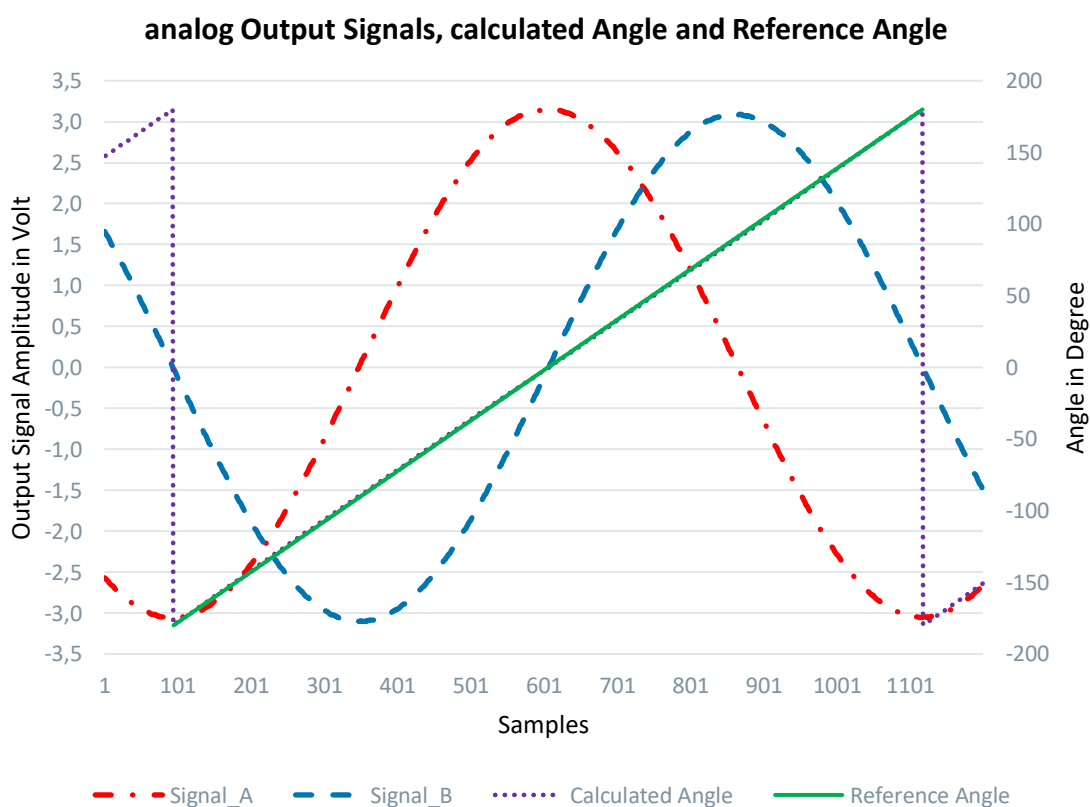
The resulting angle is illustrated in the diagram above. It ranges from -180° to +180°.

4 Integral Non-Linearity

The integral non-linearity, short INL, describes the deviation of the calculated angle from the best line of fit.

4.1 Reference Angle

Figure 8:
Analog Output Signals, Calculated Angle and Reference Angle



The best line of fit, here also called “Reference Angle” is illustrated in the diagram above.

4.2 INL Calculation

The first step when calculating the INL is, to build the difference between the calculated angle and the reference angle.

Equation 7: Difference between the actual and ideal angle

$$\Delta\varphi = \varphi_{Calc} - \varphi_{Ref}$$

with

φ_{Calc} ... *Calculated Angle*

φ_{Ref} ... *Reference Angle*

As a next step, the INL can be determined out of the maximum negative and the maximum positive deviation from the reference angle, according the following formula.

Equation 8: INL Calculation

$$INL = \frac{\Delta\varphi_{Max} - \Delta\varphi_{Min}}{2}$$

with

$\Delta\varphi_{Min}$... *maximum negative deviation from reference angle*

$\Delta\varphi_{Max}$... *maximum positive deviation from reference angle*

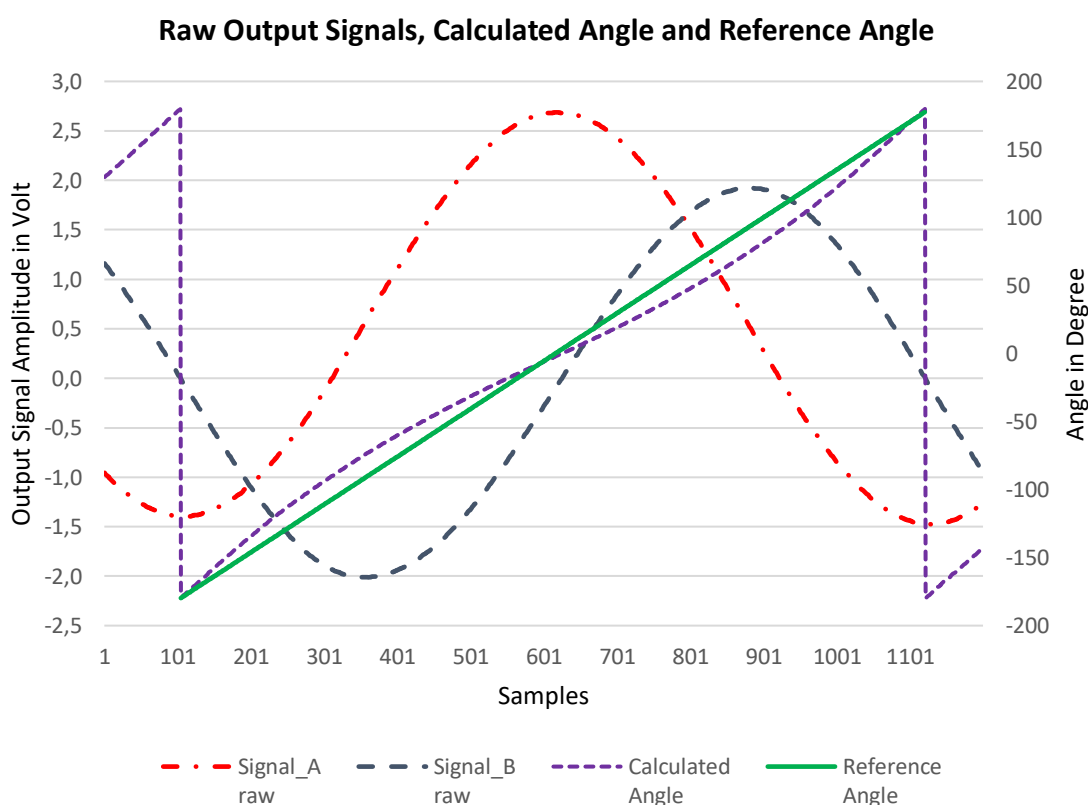
5 INL Comparison

The following Chapter clarifies how the INL improves, if it is calculated out of the gain and offset corrected signals rather than the raw signals. The calculations shown hereafter are based on actual measurement results. The complete raw data, as well as all calculations, can be found in the document “*Post_Processing_INL_Comparison.xlsx*”.

5.1 Raw Signals and corresponding Angle

Figure 9:

Output Signals before Gain and Offset Correction



As shown in Figure 9, the angle, which results from the raw output signals, is far off an ideal straight line. This will also have negative impact on the INL.

5.1.1 Calculation of INL_Raw

Equation 9: $\Delta\varphi_{Raw_Min}$ and $\Delta\varphi_{Raw_Max}$

$\Delta\varphi_{Raw,Min} = -18.5201^\circ$... maximum negative deviation from reference angle

$\Delta\varphi_{Raw,Max} = +17.2392^\circ$... maximum positive deviation from reference angle

Equation 10: INL_Raw

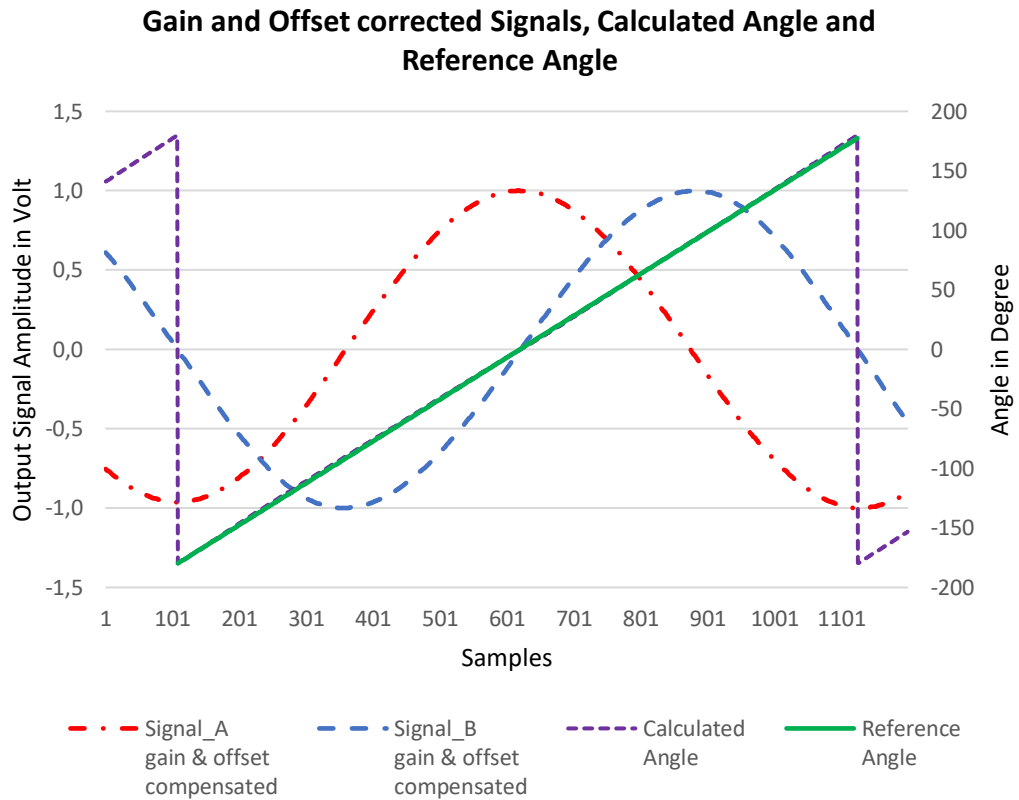
$$INL_{Raw} = \frac{\Delta\varphi_{Raw,Max} - \Delta\varphi_{Raw,Min}}{2}$$

$$INL_{Raw} = \frac{17.2392^\circ - (-18.5201^\circ)}{2} \triangleq \frac{35.7593^\circ}{2}$$

$$INL_{Raw} = 17.8796^\circ$$

5.2 Gain and Offset corrected Signals and corresponding Angle

Figure 10:
Output Signals after Gain and Offset Correction



As shown in figure 10, the angle, which results from the gain and offset corrected signals, nearly matches the reference angle.

5.2.1 Calculation of INL_Corrected

Equation 11: $\Delta\phi_{\text{Corr_Min}}$ and $\Delta\phi_{\text{Corr_Max}}$

$\Delta\phi_{\text{Corr_Min}} = -1.0896^\circ \dots$ maximum negative deviation from reference angle

$\Delta\phi_{\text{Corr_Max}} = +2.4989^\circ \dots$ maximum poitive deviation from reference angle

Equation 12: Calculation of INL_Corr

$$INL_{Corr} = \frac{\Delta\varphi_{Corr,Max} - \Delta\varphi_{Corr,Min}}{2}$$

$$INL_{Corr} = \frac{2.4989^{\circ} - (-1.0896^{\circ})}{2} \triangleq \frac{3.5885^{\circ}}{2}$$

$$INL_{Corr} = 1.7943^{\circ}$$

6 Revision Information

Changes from previous version to current revision v0-01	Page
V0-01 Initial Version	all

- Page and figure numbers for the previous version may differ from page and figure numbers in the current revision.
- Correction of typographical errors is not explicitly mentioned.

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